

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. R5-2002-0168

WASTE DISCHARGE REQUIREMENTS

FOR
AMADOR WATER AGENCY
WILDWOOD ESTATES HOMEOWNERS
WILDWOOD ESTATES COMMUNITY LEACHFIELD SYSTEM
AMADOR COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Regional Board) finds that:

1. On 13 November 2001, Amador Water Agency submitted a Report of Waste Discharge (RWD) for an existing wastewater treatment facility (WWTF) that treats and disposes of domestic wastewater generated by homes in Wildwood Estates in Amador County. Amador Water Agency shall serve as the authorized public agency (California Water Code section 13282) that ensures that the septic systems described herein are adequately designed, located, sized, spaced, constructed, and maintained to assure compliance with terms of this Order.
2. Wildwood Estates is a 40 acre residential subdivision with 37 half-acre residential parcels. The community is currently 86 percent developed; 32 of the 37 residential parcels are improved with single-family dwellings, and the remaining five are as yet unimproved.
3. Each parcel listed in Finding No. 6 has or will have a 1,500-gallon septic tank and sewage transport line owned by the parcel owner (the 37 owners of the individual parcels and septic tanks are hereafter referenced collectively and individually as "Wildwood Estate Homeowners"). Each entity shall hereafter be referred to individually "Amador Water Agency" or "Wildwood Estate Homeowners". Except when referred to individually, "Amador Water Agency" and "Wildwood Estate Homeowners" shall be referred to jointly as "Discharger." Wildwood Estate Homeowners shall be responsible for compliance with these Waste Discharge Requirements as it applies to each entity's own parcel(s) of land, while Amador Water Agency shall be responsible for assuring that design, operation, and maintenance of the entire community treatment system satisfies the terms of this Order, as described in the two "Hereby Ordered" sections of this Order.
4. For the purposes of this Order, the WWTF shall mean the septic tanks, effluent collection system (commencing at the septic tank effluent outlet) and community leachfields.
5. The community leachfields are at 12970 Burnt Cedar Lane, Amador, in Section 10, T6N, R12E MDB&M. The facility location is shown on Attachment A, which is attached hereto and made part of this Order by reference. Amador Water Agency owns and operates the collection system and community leachfield which is on Assessors Parcel Number 38-630-040.
6. Components of the WWTF are located on numerous parcels. The Wildwood Estate Homeowners own and operate their individual septic tanks. A list of the Parcel Numbers for the Wildwood Estate

Homeowners is provided below. This Order applies to the current parcel owners, as well as to any future owners of the parcels listed below.

<u>Assessor Parcel No.</u>	<u>Assessor Parcel No.</u>	<u>Assessor Parcel No.</u>	<u>Assessor Parcel No.</u>
38-630-001	38-630-011	38-630-021	38-630-047
38-630-002	38-630-012	38-630-044	38-630-032
38-630-003	38-630-013	38-630-045	38-630-033
38-630-004	38-630-014	38-630-024	38-630-034
38-630-005	38-630-015	38-630-025	38-630-035
38-630-006	38-630-016	38-630-026	38-630-036
38-630-007	38-630-017	38-630-027	38-630-037
38-630-008	38-630-018	38-630-028	
38-630-009	38-630-019	38-630-029	
38-630-010	38-630-020	38-630-030	

Existing Facility and Discharge

7. The septic tanks in the subdivision are two chamber fiberglass septic tank. The first chamber is utilized for solids removal and storage. A overflow baffle connects the first and second chambers and buffers peak hydraulic loads to the system. The secondary compartment is fitted with an effluent filter. The filtered effluent flows via gravity from the septic tank to the collection system. Effluent within the collection system system operates via gravity to supply two dosing tanks which in turn discharge the wastewater to the community leachfields.
8. The two dosing siphons and tanks are identical, but each doses a separate leachfield. The siphons used are Orenco Systems Model No. 442 and the dosing tanks are 1,500 gallon concrete septic tanks. The RWD indicates that each leachfield dose is approximately 1,200 gallons. Dosing occurs four times per day at design flow.
9. The two community leachfields consist of approximately of 4,300 lineal feet of pressure dosed leach line on a 4.3-acre parcel. The 4,300 feet of leachline is divided in approximately equal amounts between two separate leachfields. Distribution piping within each disposal trench within a leachfield is fitted with a ball valve for the purpose of equal distribution of wastewater to all the leachlines despite varying heads due to elevation differences and friction losses. These valves were adjusted at the time of the initial system testing to provide equal pressure in all disposal trenches, but may need to be checked and adjusted periodically. The existing leachfield area contains enough unused space for the replacement of 4,300 feet of leachline. The community leachfields are shown on Attachment B, which is attached hereto and made part of this Order by reference.
10. To minimize soil saturation during the wet season, the leachfield has a stormwater diversion ditch which directs stormwater away from the leachfield. The stormwater ditch is part of the WWTF and needs to be maintained in order to ensure the long term performance of the leachfield.

11. The RDW shows that the design flow for the leachfields is approximately 10,300 gallons per day.
12. Percolation tests performed on the leachfield indicate that the average percolation at a soil depth of 36 inches is approximately 39 minutes per inch and around 60 minutes per inch at a soil depth of 60 inches. The disposal trenches are 46 inches deep and 36 inches wide with 30 inches of drainage rock below the distribution lateral. The wastewater application rate is approximately 0.3 gallons per day per square foot of disposal trench.
13. On 2 July and 10 July 2001, Amador Water Agency collected samples of the wastewater effluent discharged from the two dosing siphon tanks to the leachfield. The sample data from each monitoring event are presented below.

<u>Constituent</u>	<u>Units</u>	<u>2 July 2001 Sample Concentration</u>	<u>10 July 2001 Sample Concentration</u>
pH	S.U.	6.8	6.8
Total Dissolved Solids	mg/l	282	304
Total Kjeldahl Nitrogen	mg/l	56	58
BOD ₅	mg/l	106	176
Total Coliform Organisms	MPN/100 ml	>2400	>2400
Total Fecal Organisms	MPN/100 ml	>2400	>2

14. The sample data indicates that the septic tanks provide primary treatment. Wastes that pass through the septic tanks are discharged to the soil underlying the leach lines. The soil underlying the leach lines treats some of the remaining wastes. However, the amount of treatment depends on the waste type and concentrations, soil type and depth, percolation rates, wastewater application rates, and depth to groundwater. Without adequate treatment, wastes in the wastewater may migrate through the soil column to the underlying groundwater. Therefore, this Order requires Amador Water Agency to conduct groundwater monitoring around the leachfield. Groundwater monitoring is appropriate and necessary to monitor the effectiveness of treatment within the soil column and assure groundwater quality consistent with terms specified herein.
15. Wildwood Estates Community Leachfield was formerly regulated by Amador County Environmental Health Department.
16. On 1 February 1996, the Amador County Environmental Health Department conducted an inspection of Wildwood Estates Community Leachfield. The inspection report states *"In the southeast portion of the site, near a dosing tank, numerous small cedars, less than 12" tall, were seen. Experience has shown that one of the biggest threats to leachfield longevity is infiltration of leach rock by tree roots."*
17. On 19 March 1998, the Amador County Environmental Health Department conducted a second inspection of the Wildwood Estates Community Leachfield. The inspection report states *"Several*

cedar and pine saplings have sprouted over the drainage field. Removal of these saplings should be prioritized to prevent root intrusion which is a major cause of premature leachfield failure.”

18. On 24 February 2000, Amador County Environmental Health Department conducted a third inspection of the Wildwood Estates Community Leachfield. The inspection report states “It appears that there is a broken pipe at valve box serving trenches 28, 29, 30, 46, 47, and 48. There was effluent on the ground surface and the valve box was full of effluent...No effluent was noted flowing offsite.”
19. Monitoring of the wastewater depth in the disposal trenches can aid in early detection of leachfield failure. The Amador Water Agency has installed these observation wells in the disposal trenches for this purpose. The February 2000 inspection found ponded effluent in most of the 48 disposal trenches. The depth of effluent in the disposal trenches ranged from 23 to 42 inches below ground surface (or 1 to 30 inches above the invert of the trench). Trenches range in depth from 41 to 53 inches below the ground surface.
20. An operation and maintenance manual, “Wildwood Estates Sewerage System”, was developed for the WWTF when it was originally built in 1990; however, the manual does not address operation and maintenance that are necessary to assure compliance with this Order and must be revised.

Sanitary Sewer System

21. The Wildwood Estates Community Leachfield sanitary sewer system consists of a network of two inch Schedule 40 PVC pipe laid at variable grades throughout the subdivision. Effluent from the septic tanks is directed into three primary collection mains that flow into two dosing tanks. Unlike contemporary collection systems, this collection system does not contain any manholes. Instead, air relief/vacuum valves have been installed at high points and dead end laterals. The flow rate for the collection main lines is approximately 80 gallons per minute. Peak flow design for the collection was based on a single family dwelling flow of 263 gallons per day. Wastewater flows by gravity to the dosing tanks located at the treatment plant. The entire WWTF flows via gravity and does not contain any electrically operated equipment.
22. A “sanitary sewer overflow” is defined as a discharge to ground or surface water from the sanitary sewer system at any point upstream of the wastewater treatment plant (in this case, the dosing tanks). Temporary storage and conveyance facilities (such as wet wells, regulated impoundments, tanks, highlines, etc.) may be part of a sanitary sewer system and discharges to these facilities are not considered sanitary sewer overflows, provided that the waste is fully contained within these temporary storage/conveyance facilities.
23. Within this system, a sanitary sewer overflow consists of domestic sewage. The chief causes of sanitary sewer overflows include grease blockages, root blockages, debris blockages, sewer line flood damage, air relief/vacuum valve failures, vandalism, pump station mechanical failures, power outages, storm or groundwater inflow/infiltration, lack of capacity, and contractor caused blockages.

24. Sanitary sewer overflows often contain high levels of suspended solids, pathogenic organisms, toxic wastes, nutrients, oxygen demanding organic compounds, oil and grease, and other wastes. Sanitary sewer overflows can cause temporary exceedences of applicable water quality objectives, pose a threat to public health, adversely affect aquatic life, and impair the public recreational use and aesthetic enjoyment of surface waters in the area.
25. The Amador Water Agency is expected to take all necessary steps to adequately maintain, operate, and prevent discharges from its sanitary sewer collection system. This Order requires the Amador Water Agency to prepare and implement a Sanitary Sewer System Operation, Maintenance, Overflow Prevention, and Response Plan.

Planned Changes in Discharge

26. The WWTF was designed for full buildout of the community. Therefore, the Amador Water Agency does not have plans for improvement to the WWTF at this time.

Site-Specific Conditions

27. The mean annual rainfall, based on data from Electra Power House CIMIS weather station, is 28 inches per year.
28. All portions of the WWTF are outside the 100-year flood zone.
29. The facility lies within the Middle Sierra Hydrologic Unit Area No. 532.40, as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
30. Soils on the site are of the Josephine and Sites series of metasedimentary origin. Soil depth is shallow and the depth to bedrock is approximately eight feet.

Groundwater Degradation

31. State Water Resources Control Board (SWRCB) Resolution No. 68-16 (hereafter Resolution 68-16 or the "Antidegradation Policy") requires the Regional Board in regulating the discharge of waste to maintain high quality waters of the state (i.e., background water quality) until it is demonstrated that any change in quality will be consistent with maximum benefit to the people of the State, will not unreasonably affect beneficial uses, and will not result in water quality less than that described in the Regional Board's policies (e.g., quality that exceeds water quality objectives).
32. The Regional Board finds that some degradation of groundwater beneath the leachfield is consistent with Resolution 68-16 provided that:
 - a. The degradation is confined within a specified boundary;

- b. The Amador Water Agency minimizes the degradation by fully implementing, regularly maintaining, and optimally operating best practicable treatment and control (BPTC) measures;
 - c. The degradation is limited to waste constituents typically encountered in municipal wastewater as specified in the groundwater limitations in this Order; and
 - d. The degradation does not result in water quality less than that prescribed in the Basin Plan.
33. Some degradation of groundwater by some of the typical waste constituents released with discharge from a wastewater utility after effective source control, treatment, and control may be consistent with maximum benefit to the people of California. The energy and waste management advantages of publicly managed utility service exceed individual wastewater systems. Degradation of groundwater by certain constituents (e.g., toxic chemicals) and by other constituents that can be effectively removed by conventional treatment (e.g., total coliform bacteria) is not consistent with Resolution 68-16. When allowed, the degree of degradation permitted depends upon many factors (i.e., background water quality, the waste constituent, the beneficial uses and most stringent water quality objective, source control measures, waste constituent treatability).

Treatment and Control Practices

34. This wastewater treatment facility provides treatment and control of the discharge that incorporates:
- a. An operation and maintenance (O&M) manual; and
 - b. Staffing to assure proper operation and maintenance.
35. The facility treats wastewater to primary standards and discharges the wastewater subsurface via leachlines. Because of the shallow soils, there is minimal potential for constituent attenuation in the vadose zone. The actual depth to groundwater for the site is not known. In addition, the potential impacts on groundwater and the appropriate level of degradation that complies with Resolution 68-16 have not been fully evaluated. Therefore, the Amador Water Agency's current effort may not constitute BPTC as intended in Resolution 68-16, and this Order establishes a schedule for tasks to evaluate BPTC for each conveyance, treatment, storage, and disposal component of the facility and to further characterize groundwater for selected constituents. Completion of these tasks, and implementation of the approved strategies developed from that work, will ensure that BPTC and the highest water quality consistent with the maximum benefit to the people of the State will be achieved.
36. This Order establishes interim groundwater limitations for the WWTF that will not unreasonably threaten present and anticipated beneficial uses or result in groundwater quality that exceeds water quality objectives set forth in the Basin Plan. This Order contains tasks for assuring that BPTC and the highest water quality consistent with the maximum benefit to the people of the State will be achieved. Accordingly, the discharge is consistent with the antidegradation provisions of Resolution 68-16. Based on the results of the scheduled tasks, the Regional Board may reopen this Order to reconsider groundwater limitations and other requirements to comply with Resolution 68-16.

Basin Plan, Beneficial Uses, and Regulatory Considerations

37. The *Water Quality Control Plan for the Sacramento River and San Joaquin River Basins, Fourth Edition*, (hereafter Basin Plan) designates beneficial uses, establishes water quality objectives, contains implementation plans and policies for protecting waters of the basin, and incorporates by reference plans and policies adopted by the State Water Resources Control Board. These requirements implement the Basin Plan.
38. Surface water drainage to Grass Valley Creek, tributary to Sutter Creek, tributary to Dry Creek, which is in turn tributary to the Sacramento-San Joaquin Delta.
39. The beneficial uses of Sacramento-San Joaquin Delta are municipal and domestic supply; agricultural irrigation and stock watering; industrial process and service supply; contact and other noncontact recreation; warm and cold freshwater habitat; migration for warm and cold water species; spawning for warm; wildlife habitat; and navigation.
40. The beneficial uses of the underlying groundwater are municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.
41. The Basin Plan establishes numerical and narrative water quality objectives for surface and groundwater within the basin, and recognizes that water quality objectives are achieved primarily through the Regional Board's adoption of waste discharge requirements and enforcement orders. Where numerical water quality objectives are listed, these are limits necessary for the reasonable protection of beneficial uses of the water. Where compliance with narrative water quality objectives is required, the Regional Board will, on a case-by-case basis, adopt numerical limitations in orders, which will implement the narrative objectives to protect beneficial uses of the waters of the state.
42. The Basin Plan identifies numerical water quality objectives for waters designated as municipal supply. These are the maximum contaminant levels (MCLs) specified in the following provisions of Title 22, California Code of Regulations: Tables 64431-A (Inorganic Chemicals) and 64431-B (Fluoride) of Section 64431, Table 64444-A (Organic Chemicals) of Section 64444, and Table 64449-A (Secondary Maximum Contaminant Levels-Consumer Acceptance Limits) of Section 64449. The Basin Plan's incorporation of these provisions by reference is prospective, and includes future changes to the incorporated provisions as the changes take effect. The Basin Plan recognizes that the Regional Board may apply limits more stringent than MCLs to ensure that waters do not contain chemical constituents in concentrations that adversely affect beneficial uses.
43. The Basin Plan contains narrative water quality objectives for chemical constituents, tastes and odors, and toxicity. The toxicity objective requires that groundwater be maintained free of toxic substances in concentrations that produce detrimental physiological responses in humans, plants or animals. The chemical constituent objective requires that groundwater shall not contain chemical constituents in concentrations that adversely affect beneficial uses. The tastes and odors objective

requires that groundwater shall not contain tastes or odors producing substances in concentrations that cause nuisance or adversely affect beneficial uses.

44. Section 13241 of the Water Code requires the Regional Regional Board to consider various factors, including economic considerations, when adopting water quality objectives into its Basin Plan. Water Code Section 13263 requires the Regional Board to address the factors in Section 13241 in adopting waste discharge requirements. The State Board, however, has held that a Regional Board need not specifically address the Section 13241 factors when implementing existing water quality objectives in waste discharge requirements because the factors were already considered in adopting water quality objectives. These waste discharge requirements implement adopted water quality objectives. Therefore, no additional analysis of Section 13241 factors is required.
45. The United States Environmental Protection Agency (EPA) has promulgated biosolids reuse regulations in 40 CFR 503, *Standard for the Use or Disposal of Sewage Sludge*, which establishes management criteria for protection of ground and surface waters, sets application rates for heavy metals, and establishes stabilization and disinfection criteria.
46. The Regional Board is using the Standards in 40 CFR 503 as guidelines in establishing this Order, but the Regional Board is not the implementing agency for 40 CFR 503 regulations. The Amador Water Agency may have separate and/or additional compliance, reporting, and permitting responsibilities to the EPA.
47. The State Water Resources Control Board adopted Order No. 97-03-DWQ (General Permit No. CAS000001) specifying waste discharge requirements for discharges of storm water associated with industrial activities, and requiring submittal of a Notice of Intent by all affected industrial dischargers. The wastewater treatment plant facilities are located underground and are not exposed to stormwater runoff. Because there is no storm water discharge, the Amador Water Agency is not required to obtain coverage under General Permit No. CAS000001.
48. On 7 February 1990, in accordance with the California Environmental Quality Act (CCR, Title 14, Section 15261 et. seq.), the Amador County Planning Commission certified a final Negative Declaration for subdivision project including the community wastewater collection and leachfield disposal system.
49. Section 13267(b) of California Water Code provides that: *“In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the*

regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports.”

The technical reports required by this Order and the attached “Monitoring and Reporting Program No. ____” are necessary to assure compliance with these waste discharge requirements. The Discharger operates facilities that discharge waste subject to this Order.

50. The California Department of Water Resources sets standards for the construction and destruction of groundwater wells (hereafter DWR Well Standards), as described in *California Well Standards Bulletin 74-90* (June 1991) and *Water Well Standards: State of California Bulletin 94-81* (December 1981). These standards, and any more stringent standards adopted by the state or county pursuant to CWC Section 13801, apply to all monitoring wells.
51. State regulations that prescribe procedures for detecting and characterizing the impact of waste constituents from waste management units on groundwater are found in Title 27. While the wastewater treatment facility is exempt from Title 27, the data analysis methods of Title 27 may be appropriate for determining whether the discharge complies with the terms for protection of groundwater specified in this Order.
52. The discharge authorized herein and the treatment and storage facilities associated with the discharge, except for discharges of residual sludge and solid waste, are exempt from the requirements of Title 27, California Code of Regulations (CCR), Section 20380 et seq. (hereafter Title 27). The exemption, pursuant to Title 27 CCR Section 20090(a), is based on the following:
 - a. The waste consists primarily of domestic sewage and treated effluent;
 - b. The waste discharge requirements are consistent with water quality objectives; and
 - c. The treatment and storage facilities described herein are associated with a municipal wastewater treatment plant.
53. Pursuant to California Water Code Section 13263(g), discharge is a privilege, not a right, and adoption of this Order does not create a vested right to continue the discharge.

Public Notice

54. The Regional Board considered all the above and the supplemental information and details in the attached Information Sheet, which is incorporated by reference herein, in establishing the following conditions of discharge.
55. The Regional Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge, and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
56. The Regional Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that pursuant to Sections 13263 and 13267 of the California Water Code, Wildwood Estates Homeowners, their agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted hereunder, shall comply with the following on their respective properties:

[Note: Other prohibitions, conditions, definitions, and some methods of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated 1 March 1991.]

A. Discharge Prohibitions

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Surfacing of wastewater around the septic tank is prohibited.
4. Discharge of waste classified as 'hazardous' under Section 2521, Chapter 15 of Title 23 or 'designated', as defined in Section 13173 of California Water Code is prohibited.

B. Discharge Specifications and Provisions

1. Wastewater storage and treatment shall not cause pollution or a nuisance as defined by Section 13050 of the California Water Code.
2. No waste constituent shall be released or discharged, or placed where it will be released or discharged, in a concentration or in a mass that causes violation of the Groundwater Limitations.
3. All systems and equipment shall be maintained and operated to maximize treatment of wastewater and optimize the quality of the discharge. In particular, the Wildwood Estate Homeowners shall properly maintain the sewage piping and septic tanks including the timely removal and disposal of accumulated scum/sludge. Septic tanks that are cracked or otherwise damaged shall be promptly repaired or replaced.
4. Sludge and solid waste removed from septic tanks shall be pumped and haul only by a licensed septage hauler.
5. The Wildwood Estate Homeowners shall comply with the Monitoring and Reporting Program No. _____, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
6. The Wildwood Estate Homeowners shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are

attached hereto and made part of this Order by reference. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."

7. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

IT IS HEREBY ORDERED that pursuant to Sections 13263 and 13267 of the California Water Code, Amador Water Agency, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted hereunder, shall comply with the following:

[Note: Other prohibitions, conditions, definitions, and some methods of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated 1 March 1991.]

C. Discharge Prohibitions

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of sewage from a sanitary sewer system at any point upstream of the dosing tanks is prohibited. Discharge of treated wastewater outside of the leachfield area is prohibited.
4. Surfacing of wastewater within or downgradient of the leachfield is prohibited.
5. Discharge of waste classified as 'hazardous' under Section 2521, Chapter 15 of Title 23 or 'designated', as defined in Section 13173 of California Water Code is prohibited.
6. The discharge of wastewater to any leachfield trench where a standpipe shows wastewater is within 12 inches of ground surface is prohibited.

D. Discharge Specifications

1. The 30-day average flow shall not exceed 10,300 gallons per day.
2. Wastewater treatment and leachfield area shall not cause pollution or a nuisance as defined by Section 13050 of the California Water Code .
3. Public contact with wastewater shall be precluded or controlled through such means as fences and signs, or acceptable alternatives.

4. No waste constituent shall be released or discharged, or placed where it will be released or discharged, in a concentration or in a mass that causes violation of the Groundwater Limitations.
5. Objectionable odor originating at the leachfield shall not be perceivable beyond the limits of the property owned by the Amador Water Agency.
6. All treatment, storage, and leachfields shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
7. Application of wastewater shall be confined to the leachfields as defined in this Order.
8. The Amador Water Agency shall operate all systems and equipment to maximize treatment of wastewater and optimize the quality of the discharge. In particular,
 - a. The Amador Water Agency shall conduct inspections of the septic tanks on at least an annual basis.
 - b. The Amador Water Agency shall cut the vegetation and remove the plant litter from the leachfield as needed to prevent root intrusion into the leachlines and drainage rocks.
 - c. The Amador Water Agency shall periodically evaluate the distribution of wastewater to all the leachlines and make modifications to the systems as necessary.
 - d. The Amador Water Agency shall semi-annually evaluate the stormwater diversion ditch and shall maintain the ditch as necessary, including controlling vegetation.
 - e. The Amador Water Agency shall operate and maintain the WWTF in accordance with the Operation and Maintenance Plan specified in Provision No. H.1.c.
9. A 100-foot buffer zone shall be maintained between any watercourse and the leachfield.
10. A 100-foot buffer zone shall be maintained between any spring, domestic well or irrigation well and the leachfield.
11. A 50-foot buffer zone shall be maintained between the leachfield and all property boundaries.

E. Effluent Limitations

Effluent discharged from the dosing tanks shall not have a pH less than 6.5 or greater than 8.5 .

F. General Solids Disposal Specifications

Sludge means the solid, semisolid, and liquid residues removed during primary, secondary, or advanced wastewater treatment processes. Solid waste refers to grit and screenings generated during preliminary treatment. Residual sludge means sludge that will not be subject to further treatment at the facility. Biosolids refers to sludge that has undergone sufficient treatment and testing to qualify for reuse pursuant to federal and state regulations as a soil amendment for agriculture, silviculture, horticulture, and land reclamation.

1. Sludge and solid waste shall be removed from screens, septic tanks, and dosing tanks as needed to ensure optimal plant operation.
2. Treatment and storage of sludge shall be confined to the treatment facility property, and shall be conducted in a manner that precludes infiltration of waste constituents into soils in a mass or at concentrations that will violate the Groundwater Limitations of this Order.
3. Any storage of residual sludge, solid waste, and biosolids at the facility shall be temporary, and the waste shall be controlled and contained in a manner that minimizes leachate formation and precludes infiltration of waste constituents into soils in a mass or at concentrations that will violate the Groundwater Limitations of this Order.
4. Residual sludge, biosolids, and solid waste shall be disposed of in a manner approved by the Executive Officer and consistent with Title 27. Removal for further treatment, disposal, or reuse at disposal sites operated in accordance with valid waste discharge requirements issued by a regional water quality control board will satisfy this specification.
5. Use and disposal of biosolids should comply with the self-implementing federal regulations of 40 CFR 503, which are subject to enforcement by the U.S. EPA, not the Regional Board. If during the life of this Order, the State accepts primacy for implementation of 40 CFR 503, then the Regional Board may also initiate enforcement where appropriate.

G. Groundwater Limitations

1. Release of waste constituents from the leachfields shall not cause groundwater, as determined by an approved well monitoring network, to:
 - a. Contain any of the following constituents in concentration greater than as listed or greater than ambient background quality, whichever is greater:

<u>Constituent</u>	<u>Units</u>	<u>Limitation</u>
Boron	mg/L	0.6
Chloride	mg/L	106
Iron	mg/L	0.3
Manganese	mg/L	0.05

<u>Constituent</u>	<u>Units</u>	<u>Limitation</u>
Sodium	mg/L	69
Total Coliform Organisms	MPN/100 mL	Nondetect
Total Dissolved Solids ^a	mg/L	450
Total Nitrogen	mg/L	10
Nitrite (as N)	mg/L	1
Nitrate (as N)	mg/L	10
Ammonia (as N)	mg/l	0.5

- a. A cumulative impact limit that accounts for several dissolved constituents in addition to those listed here separately [e.g., alkalinity (carbonate and bicarbonate), calcium, hardness, phosphate, and potassium].
- b. Contain any constituent not identified in Groundwater Limitation G.1.a in concentrations greater than background quality (whether chemical, physical, biological, bacteriological, radiological, or some other property or characteristic).
- c. Exhibit a pH of less than 6.5 or greater than 8.5 pH units.
- d. Impart taste, odor, toxicity, or color that creates nuisance or impairs any beneficial use.

H. Provisions

1. All of the following reports shall be submitted pursuant to Section 13267 of the California Water Code and shall be prepared by a registered professional as described by Provision H.3.
 - a. By **1 September 2002**, the Amador Water Agency shall submit a *Revenue Plan* that describes the costs associated with completing Provisions Numbers H.1.c. through H.1.j. and shows how the Amador Water Agency will finance each item. Should the Revenue Plan show that there are inadequate funds, the Amador Water Agency must also include an implementation schedule that shows how the Amador Water Agency will raise the necessary funds in time to comply with the appropriate due dates.
 - b. **Within 120 days of approval** of the Revenue Plan by the Executive Officer, the Amador Water Agency shall submit evidence of adoption of the Revenue Plan.
 - c. By **1 September 2002**, the Amador Water Agency shall submit and implement an Operation and Maintenance (O&M) Plan for the wastewater treatment facility and leachfield. The O&M Plan shall instruct field personnel on how to manage the day-to-day discharge operations to comply with the terms and conditions of this Order and how to make field adjustments, as necessary, to preclude nuisance conditions (e.g., surfacing water). It shall also include a nuisance condition troubleshooting flowchart and a description of notification requirements. A copy of the O&M Plan shall be kept at the

facility for reference by operating personnel. Key personnel shall be familiar with its contents. The O&M Plan shall include the following documents as report appendices:

- i. A Vegetation Control Plan which describes how vegetation will be controlled and maintained in the leachfield to prevent root intrusion and damage.
- ii. A Septic Tank Inspection Plan which describes the procedures for annually testing septic tanks to determine if the septic tank requires the collected scum/sludge to be removed and description of the inspection procedures for checking the intergity septic tank.
- iii. A description of the type and location of the flow meter which has been installed to meet compliance with the Monitoring and Reporting Program.

SANITARY SEWER EVALUATION

- d. By **1 September 2002**, the Amador Water Agency shall submit a *Sanitary Sewer System Operation, Maintenance, Overflow Prevention, and Response Plan* (SSS Plan) that describes the actions designed to prevent, or minimize the potential for sanitary sewer overflows. The Amador Water Agency shall maintain the SSS Plan in an up-to-date condition and shall amend the SSS Plan whenever there is a change (e.g. in the design, construction, operation, or maintenance of the sanitary sewer system or sewer facilities) that materially affects the potential for sanitary sewer overflows, or whenever there is a sanitary sewer overflow. The Amador Water Agency shall ensure that the up-to-date SSS Plan is readily available to sewer system personnel at all times and that sewer system personnel are familiar with it.
 - i. At a minimum, the Operation and Maintenance portion of the plan shall contain or describe the following:
 1. Detailed maps of the sanitary sewer system, identifying sewer mains, air relief valves, and lift stations;
 2. A detailed listing of elements to be inspected, a description of inspection procedures and inspection frequency, and sample inspection forms;
 3. A schedule for routine inspection and testing of all pipelines, lift stations, valves, and other key system components. The inspection/testing program shall be designed to reveal problems that might lead to accidental spills and ensure that preventive maintenance is completed;
 4. Provisions for repair or replacement of old, worn out, or defective equipment;
 5. Provisions to minimize the need for manual operation of critical systems and provide spill alarms or other “fail safe” mechanisms;

6. The ability to properly manage, operate and maintain, at all times, all parts of the collection system that the Amador Water Agency owns or over which the Amador Water Agency has operational control;
 7. The ability to provide adequate capacity to convey base flows and peak flows for all parts of the collection system the Amador Water Agency owns or over which the Amador Water Agency has operational control; and
 8. How the Amador Water Agency will take all feasible steps to stop and mitigate the impact of sanitary sewer overflows in portions of the collection system the Amador Water Agency owns or over which the Amador Water Agency has operational control.
- ii. At a minimum, the Overflow Prevention and Response Plan shall contain or describe the following:
1. Identification of areas of the collection system that historically have overflowed and an evaluation of the cause of the overflow;
 2. Maintenance activities that can be implemented to address the cause of the overflow and means to prevent future overflows. Maintenance activities may include pretreatment of wastewater from industrial dischargers who discharge high concentrations of oil and grease in their wastewater;
 3. Procedures for responding to sanitary sewer overflows designed to minimize the volume of sewer overflow that enters surface waters, and minimize the adverse effects of sewer overflows on water quality and beneficial uses;
 4. Steps to be taken when an overflow or spill occurs, and procedures that will be implemented to ensure that all overflows and spills are properly identified, responded to and reported; and
 5. A public notification plan, in which any posting of areas contaminated with sewage is performed at the direction of the Amador County Environmental Health Department. All parties with a reasonable potential for exposure to an overflow event shall be notified.

GROUNDWATER QUALITY EVALUATION

- e. By **1 June 2004**, the Amador Water Agency shall submit a workplan for the characterization of groundwater quality. The workplan shall describe the installation of groundwater monitoring wells to allow evaluation of the groundwater quality upgradient and downgradient of the leachfield. Every monitoring well shall be constructed to yield representative samples from the uppermost layer of the uppermost aquifer and to comply with applicable well standards. The workplan shall be consistent with, and include the items listed in, the first section of Attachment C, *“Items to be Included in a Monitoring Well Installation Workplan and a Monitoring Well Installation Report of Results.”*

- f. By **30 August 2004**, the Amador Water Agency shall submit a groundwater well installation report. The report shall be consistent with, and include the items listed in, the second section of Attachment C. The report shall clearly show that Amador Water Agency has the expertise and equipment necessary to collect quarterly groundwater samples. Alternatively, the report may describe the qualified consultant that the Amador Water Agency will use to collect groundwater samples.
- g. By **30 December 2005**, the Amador Water Agency shall submit a *Background Groundwater Quality Study Report*. For each groundwater monitoring parameter/constituent identified in the Monitoring and Reporting Program, the report shall present a summary of monitoring data, calculation of the concentration in background monitoring wells, and comparison of background groundwater quality to that in wells used to monitor the facility. Determination of background quality shall be made using the methods described in Title 27, Section 20415(e)(10), and shall be based on data from at least four consecutive quarterly (or more frequent) groundwater monitoring events. For each monitoring parameter/constituent, the report shall compare measured concentrations for compliance monitoring wells with: 1) the calculated background concentration, and 2) the interim numeric limitations set forth in Groundwater Limitation G.1.a. Where background concentrations are statistically greater than the interim limitations specified in Groundwater Limitation G.1.a, the report shall recommend final groundwater limitations which comply with Resolution 68-16 for the waste constituents listed therein. Subsequent use of a concentration as a final groundwater limitation will be subject to the discretion of the Executive Officer.

ANTIDEGRADATION EVALUATION

- h. By **30 March 2006**, the Amador Water Agency shall submit a *BPTC Evaluation Workplan* that sets forth the scope and schedule for a systematic and comprehensive technical evaluation of each component of the facility's waste treatment and disposal system to determine best practicable treatment and control for each waste constituent listed in the Groundwater Limitation G.1.a of this Order. The workplan shall contain a preliminary evaluation of each component of the WWTF and effluent disposal system and propose a time schedule for completing the comprehensive technical evaluation. The schedule to complete the evaluation shall be as short as practicable, and shall not exceed one year.
- i. By **30 April 2007**, the Amador Water Agency shall submit a *BPTC Evaluation Report and Implementation Workplan*. The report shall include a comprehensive evaluation of the BPTC measures studied, a discussion of BPTC measures proposed for implementation (i.e., recommendations for WWTF modifications), estimated concentration or mass loading reductions for each BPTC measure, specific methods the Amador Water Agency proposes to monitor and assure continuous optimal performance of BPTC measures, the source of funding, and proposed schedule for modifications. The schedule for full implementation shall be as short as practicable, and in no case shall it exceed four years past the Executive Officer's approval of the workplan unless

specifically approved by the Regional Board. The component evaluation, recommended improvements, and implementation schedule are subject to the Executive Officer's approval.

- j. By **30 April 2009**, the Amador Water Agency shall submit a technical report that proposes specific numeric groundwater limitations that (a) reflects compliance with applicable water quality objectives, background water quality, and full implementation of BPTC measures, and (b) describes how these were determined considering actual data from compliance monitoring wells, impact reductions through full implementation of BPTC, reasonable growth, etc. The Amador Water Agency should submit results of a validated groundwater model to support its proposal. In addition, the technical report shall describe the overall status of compliance with implementation of BPTC measures and compliance with all groundwater limitations.
2. Upon completion of tasks set forth in Provision H.1, the Regional Board shall consider the evidence provided and make a determination regarding whether the Amador Water Agency has justified BPTC and the appropriate final numeric groundwater limitations that comply with Resolution 68-16.
3. In accordance with California Business and Professions Code sections 6735, 7835, and 7835.1, engineering and geologic evaluations and judgments shall be performed by or under the direction of registered professionals competent and proficient in the fields pertinent to the required activities. All technical reports specified herein that contain workplans for, that describe the conduct of investigations and studies, or that contain technical conclusions and recommendations concerning engineering and geology shall be prepared by or under the direction of appropriately qualified professional(s), even if not explicitly stated. Each technical report submitted by the Amador Water Agency shall contain a statement of qualifications of the responsible licensed professional(s) as well as the professional's signature and/or stamp of the seal.
4. The Amador Water Agency shall comply with the Monitoring and Reporting Program No. _____, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
5. The Amador Water Agency shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and made part of this Order by reference. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
6. The Amador Water Agency shall use the best practicable cost-effective control technique(s) including proper operation and maintenance, to assure compliance with terms of this Order.
7. The Amador Water Agency shall provide certified wastewater treatment plant operators in accordance with Title 23 of the California Code of Regulations, Division 3, Chapter 26.

8. As described in the Standard Provisions, the Amador Water Agency shall report promptly to the Regional Board any material change or proposed change in the character, location, or volume of the discharge.
9. Upon the reduction, loss, or failure of the sanitary sewer system resulting in a sanitary sewer overflow, the Amador Water Agency shall take any necessary remedial action to (a) control or limit the volume of sewage discharged, (b) terminate the sewage discharge as rapidly as possible, and (c) recover as much as possible of the sewage discharged (including wash down water) for proper disposal. The Amador Water Agency shall implement all applicable remedial actions including, but not limited to, the following:
 - a. Interception and rerouting of sewage flows around the sewage line failure;
 - b. Vacuum truck recovery of sanitary sewer overflows and wash down water;
 - c. Use of portable aerators where complete recovery of the sanitary sewer overflows are not practicable and where severe oxygen depletion is expected in surface waters; and
 - d. Cleanup of sewage-related debris at the overflow site.
10. The Amador Water Agency shall report to the Regional Board any toxic chemical release data it reports to the State Emergency Response Commission within 15 days of reporting the data to the Commission pursuant to section 313 of the "Emergency Planning and Community Right to Know Act of 1986."
11. The Amador Water Agency shall not allow waste-free wastewater to be discharged into the wastewater collection, treatment, and disposal system in amounts that significantly diminish the system's capability to comply with this Order. Waste-free wastewater means rainfall, groundwater, cooling waters, and condensates that are essentially free of wastes.
12. The Amador Water Agency shall submit to the Regional Board on or before each compliance report due date, the specified document or, if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is being reported, then the Discharge shall state the reasons for such noncompliance and provide an estimate of the date when the Amador Water Agency will be in compliance. The Amador Water Agency shall notify the Regional Board in writing when it returns to compliance with the time schedule.
13. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Amador Water Agency shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
14. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or recycling areas or off-site reuse of effluent, used to justify the capacity

authorized herein and assure compliance with this Order, the Amador Water Agency shall notify the Regional Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.

15. The Amador Water Agency must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or recession of this Order.
16. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
17. The Regional Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 6 September 2002.

GARY M. CARLTON, Executive Officer